

OUTPATIENT QUEUE MANAGEMENT SYSTEM (OQMS)

Business Requirements Document

Business Analyst Portfolio Project

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Domain: Healthcare

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1. Case Study Summary

Domain: Healthcare

Problem: The hospital's Outpatient Department lacked visibility and structure in its queue process. Patients waited 3-4 hours past appointment times with no status visibility, doctors were repeatedly interrupted to resolve queue questions, walk-in patients created ordering confusion, and patient files were often not ready when patients were called in all while patient privacy had to be strictly protected in a shared waiting area.

My Role: *Business Analyst*: conducted stakeholder analysis across 9 stakeholder groups, authored the Business Requirements Document, performed a 7-point As-Is/To-Be gap analysis, and defined user stories and acceptance criteria for the proposed system.

Approach: Identified competing priorities across patients, clinical staff, compliance, and IT. Deliberately scoped the solution as a queue/appointment reference layer not a medical records system to avoid unnecessary risk and complexity, while still ensuring doctors could securely reference existing patient history. Addressed two explicit non-functional requirements (data privacy and offline reliability) alongside standard process requirements.

Key Artifacts: Stakeholder Analysis, Business Requirements Document, As-Is/To-Be Gap Analysis, Process Flow Diagram, User Stories & Acceptance Criteria.

Projected Impact: 90% reduction in reception inquiries about queue status; zero display of patient personal information in public areas; 100% of patient files ready before consultation; full operational continuity during WiFi/power outages.

Skills Demonstrated: Stakeholder analysis across a complex multi-department environment, non-functional requirements engineering (privacy, reliability), deliberate scope boundary-setting, dependency/risk identification, process gap analysis, user story writing.

2. Stakeholder Analysis

Nine stakeholder groups were identified across the patient journey, clinical operations, compliance, and technical delivery, and assessed for their interest, influence, and level of support establishing the competing priorities that shaped the Business Requirements Document.

Stakeholder	Interest / Concern	Influence	Support
Scheduled Patients	Want accurate appointment timing and visibility into queue status to avoid long in-person waits.	Medium	Champion
Walk-in Patients	Want fair treatment in the queue without being indefinitely delayed behind scheduled appointments.	Low	Neutral
Elderly / Visually Impaired Patients	May struggle to use the app or read the digital display; need a reliable non-digital fallback.	Low	Cautious

Stakeholder	Interest / Concern	Influence	Support
Doctors	Need patient files ready on time and fewer queue-related interruptions during consultations.	High	Champion
Nurses / Clinical Support Staff	Responsible for file preparation need advance notice of upcoming patients to stay ahead of the queue.	High	Champion
Reception / Front Desk Staff	Currently field repeated “who's next” questions want reduced manual queue management burden.	Medium	Champion
Hospital Administration / Management	Wants improved patient satisfaction and operational efficiency without compromising compliance.	High	Champion
Data Privacy and Compliance Officer	Concerned that public queue displays could expose patient identity needs strict anonymization.	High	Cautious
IT Team	Responsible for building the app, offline sync, and EHR integration concerned with technical feasibility.	Medium	Neutral

3. Business Requirements Document

3.1 Executive Summary

Problem: The hospital's Outpatient Department currently lacks a system to manage patient queues and documentation effectively. Patients cannot see their position in the queue and often wait 3-4 hours past their appointment time. Walk-in and elderly patients create additional confusion in queue ordering, staff are repeatedly interrupted to answer “who's next,” and doctors sometimes see patients without the necessary supporting documents already prepared.

Solution: Develop a queue and appointment management system that lets patients book appointments, reminds them to bring required documents, manages queue order fairly for both scheduled and walk-in patients, allows doctors and nurses to reference existing patient records securely, and continues operating during power or network outages.

3.2 Business Objective

Enable patients to book appointments remotely and see their queue position and the current number being served, reducing unnecessary early arrival and long in-person waits. Ensure patients are reminded to bring required documents ahead of their visit. Allow doctors and nurses to securely reference existing patient medical history for informed treatment. Display queue numbers on a public digital board to manage order transparently, and ensure the system remains operational during WiFi or power interruptions.

3.3 Scope

In Scope:

- A patient-facing mobile app for appointment booking and document reminders.
- Real-time queue number visibility for patients via the app (current number being served, upcoming number).
- A digital display board at the hospital for public queue tracking.
- A fair queue-ordering process for walk-in patients alongside scheduled appointments (e.g., a 10:1 ratio).
- Secure, permissioned access for doctors and nurses to reference a patient's existing medical records for treatment purposes.
- Offline functionality allowing the queue to continue operating during WiFi or power interruptions, with automatic sync once restored.

Out of Scope:

- Electronic Health Record (EHR) management the system allows doctors and nurses to reference existing patient records via integration with the hospital's medical records system, but does not itself capture, store, or manage new clinical diagnoses or treatment data (see Assumptions).
- Billing and payment processing.

3.4 Business Requirements

ID	Category	Requirement
BR-01	Core Function	The system shall provide a patient-facing mobile app for booking appointments.
BR-02	Core Function	The system shall remind patients via the app to bring all necessary documents ahead of their appointment.
BR-03	Core Function	The system shall allow patients to view their upcoming queue number and the number currently being served, from home or in the hospital.
BR-04	Logic	The system shall interleave walk-in patients with scheduled appointments using a fair ratio (e.g., 10 appointments to 1 walk-in) to prevent indefinite walk-in wait times.
BR-05	Workflow	The system shall allow doctors and nurses to securely reference a patient's existing medical history for treatment purposes, with access limited to authorized clinical staff.
BR-06	NFR: Reliability	The system shall continue operating during WiFi or power interruptions, with queue data cached locally and automatically synced once connectivity/power is restored.
BR-07	NFR: Privacy	The system shall not display or announce any patient name or personal detail on public screens or in the waiting area only anonymized queue numbers may be shown.

3.5 Assumptions & Constraints

Assumptions:

- Patients possess smartphone access and basic digital literacy to navigate the app.
- A non-digital or assisted option (reception/nursing staff support) will remain available at the hospital for elderly patients or those with visual impairments who cannot use the app or read the digital display.
- The hospital maintains, or will maintain, a separate medical records/EHR system where doctors and nurses record diagnoses, treatment notes, and clinical history during and after consultations. OQMS integrates with this system only to retrieve existing patient records for reference it is not responsible for capturing or storing new clinical data entered during a consultation.

Constraints:

- The solution must comply with the hospital's patient privacy, data ethics, and confidentiality obligations at all times.
- No patient name, address, or other personally identifiable information may pass through the public-facing display or be announced verbally in the waiting area.
- The system must function without continuous WiFi or power availability, given known connectivity limitations in parts of the hospital building.

3.6 Risks

Risk	Description	Mitigation
Privacy breach	If the system fails entirely, staff may revert to calling out patient names to manage the queue manually, breaching confidentiality.	Maintain a manual, paper-based token backup system at reception that does not depend on power, network, or software functioning.
Digital exclusion	Elderly patients or those with visual impairments may struggle to read or use the digital queue display or app.	Reception and nursing staff will be available to guide these patients through the queue process directly.
File retrieval delay	Medical file/record retrieval may not keep pace with the new digital queue, so patients could reach the consultation room before their file is ready.	Coordinate with nursing staff to trigger file preparation once a patient is a set number of positions away, not only when called.
Dependency on external EHR system	If the hospital lacks a functioning medical records system, OQMS's reference-only scope becomes unworkable.	Confirm during requirements validation whether a medical records system exists; if not, treat it as a prerequisite project or documented dependency.

3.7 Success Metrics / KPIs

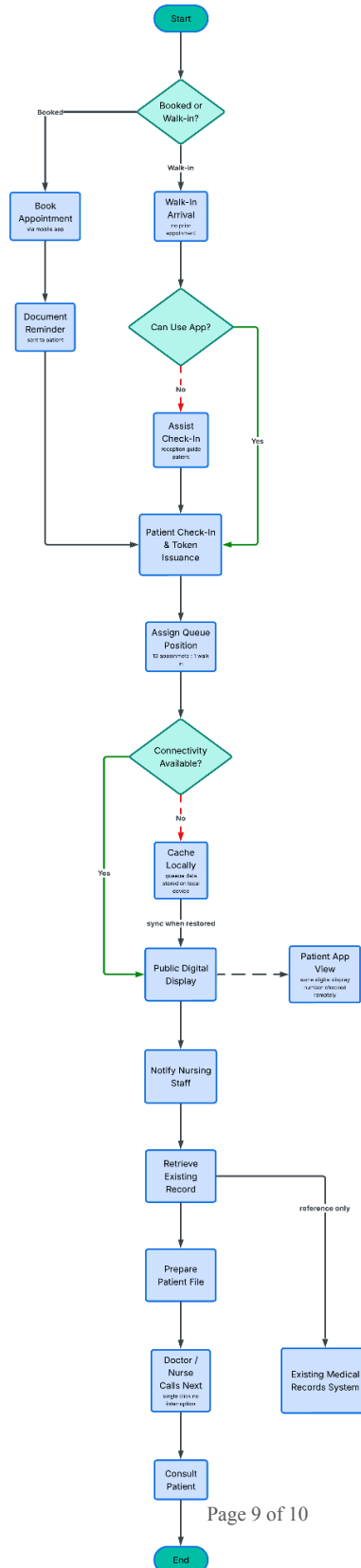
- 90% reduction in patient inquiries at reception regarding queue status or wait time.
- Zero instances of patient names or personal details displayed or announced publicly.
- 100% of medical files/records ready before the patient enters the consultation room.
- 100% operational continuity of the queue system during recorded WiFi or power outages.

4. As-Is / To-Be Gap Analysis

Process Step	As-Is (Current)	To-Be (Future)	Gap (What needs to change / be built)
1. Appointment Booking	Patients must call or visit the hospital in person to book an appointment.	Patients book appointments remotely through a mobile app.	New System: patient-facing mobile app with an appointment booking module (BR-01).
2. Document Preparedness	Patients often arrive without required documents, causing delays at consultation.	The app sends automated reminders of required documents ahead of the visit.	New Feature: automated document-reminder notifications tied to appointment type (BR-02).
3. Queue Visibility	Patients have no way to know their position and wait in person for 3-4 hours past their appointment time.	Patients view their queue number and the number currently being served, remotely and on-site.	New System: real-time queue tracking module plus a public digital display board (BR-03).
4. Walk-in vs. Scheduled Ordering	Walk-ins create manual, inconsistent ordering disputes that repeatedly interrupt staff.	The system automatically interleaves walk-ins with scheduled patients at a fair ratio (10:1).	New Logic: automated queue-sequencing / interleaving algorithm (BR-04).
5. Clinical Record Reference	Doctors sometimes see patients without the necessary supporting history or documents prepared.	Doctors and nurses securely reference existing patient records through system integration.	New Integration: secure, permissioned reference link to the existing EHR / medical records system (BR-05).
6. System Reliability	No documented contingency exists; a WiFi or power outage would halt queue management entirely.	The system caches queue data locally and automatically syncs once power/network is restored.	New Architecture: offline-first design with local caching and auto-sync (BR-06).
7. Privacy in Public Areas	Reception and clinical staff call out patient names to manage the queue, risking confidentiality breaches.	Only anonymized queue numbers are shown on the display board and referenced verbally.	New Process: privacy-by-design display logic; removal of name-based calling (BR-07).

5. Process Flow Diagram

The diagram below illustrates how scheduled and walk-in patients are both processed through token generation, fair queue interleaving, anonymized public display, and file preparation with offline resilience (BR-06) underpinning every stage, and the existing medical records system referenced but explicitly kept outside this project's scope.



6. User Stories & Acceptance Criteria

- As a patient, I want to book an appointment through a mobile app, so that I don't need to visit the hospital in person to schedule a visit.
- As a patient, I want to receive a reminder of required documents before my appointment, so that I arrive fully prepared.
- As a patient, I want to see my upcoming queue number and the number currently being served, so that I can time my arrival and avoid long unnecessary waits.
- As a walk-in patient, I want to be fairly slotted into the queue, so that I am not indefinitely delayed behind scheduled appointments.
- As a doctor, I want to securely reference a patient's existing medical history, so that I can make informed treatment decisions without delay.
- As a nurse, I want to be notified to prepare a patient's file a few positions before their turn, so that the file is ready when they are called.
- As a hospital administrator, I want assurance that no patient's personal information is displayed publicly, so that the hospital meets its confidentiality obligations.

Acceptance Criteria - Queue Visibility Story

***Given** a patient has booked or checked in for an appointment,*

***When** they open the queue tracking screen in the app,*

***Then** they see their own queue number and the number currently being served, without any other patient's personal details visible.*

Acceptance Criteria - File Prep Trigger Story

***Given** a patient is 3 queue positions away from their consultation,*

***When** the system reaches that threshold,*

***Then** a notification is sent to nursing staff to retrieve and prepare the patient's existing file for the doctor.*